4.4 Empirical Validation

We evaluate descaling on the repetition task from Section 2. Figure 7 shows results for Qwen2.5-7B-Instruct. With descaling, the model accurately follows repetition counts up to 512 (max tested), compared to uncontrolled repetition in the baseline. Entropy measurements confirm our theoretical predictions: descaling maintains higher token diversity throughout generation.

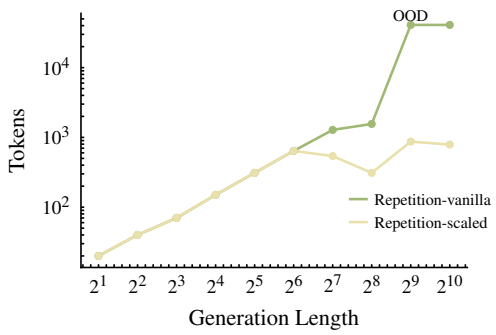

Figure 7: Maximum achieved repetition count with vs without descaling during generation (target=512 repetitions).

Alternative Scaling Functions. We experimented with linear $(1/t)$ and constant scaling (See Table 2), but found logarithmic decay offers the best sufficiently aggressive to prevent runaway repetition. Constant scaling (e.g., 0.5×) failed in the repetition control. Linear scaling causes premature collapse of induction head utility.

Scaling Function	Repetition Control
Logarithmic $(1/\log(t+c))$	118.9
Linear $(1/t)$	186.7
Constant (e.g., 0.5×)	OOD

Table 2: Comparison of alternative scaling functions for induction head descaling. Logarithmic scaling achieves the best balance between repetition control. Here repetition control refers to the final generated target token under repetition-512.

Ablation Study. As for the selection of c, we tested it on the repetition-512 test on Qwen2.5-7B-Instruct, considering four cases: (i) A small c (e.g., $c = 1$), representing relatively weak control of repetition. (ii) A large c (e.g., $c = 10$), representing relatively strong control of repetition. (iii) A negative or zero c, which disables the induction heads / writes into a contrast direction to the residual stream. (iv) An optimal c (e.g., $c = 2$), which balances repetition control and model performance. Our experiments show that a small c fails to sufficiently suppress repetition, while a large c overly restricts induction heads, impairing the model's ability to perform in-context learning tasks. Negative or zero values of c render the scaling function invalid, leading to erratic behavior. The optimal choice ($c = 2$) effectively mitigates the repetition curse without compromising the model's core capabilities.

Generation Performance. To evaluate the impact of our proposed logarithmic descaling method on generation quality, we benchmarked the perplexity of text generated by Qwen2.5-7B-Instruct under different scaling configurations (See Table 3). Perplexity measures how well the model predicts a sequence of tokens, with lower values indicating better performance. We compared the following scenarios: (i) Baseline (no descaling), (ii) Logarithmic descaling with optimal $c = 10$, (iii) Linear descaling, and (iv) Constant descaling (0.5×). The evaluation was conducted on a diverse set of text generation tasks, including narrative storytelling, technical writing, and conversational dialogue.

Method	Perplexity (PPL)
Baseline (No Descaling)	2.21
Logarithmic Descaling ($c = 2$)	2.25
Linear Descaling	2.27
Constant Descaling (0.5×)	2.31

Table 3: Perplexity scores for text generation under different descaling methods. Logarithmic descaling with $c = 2$ achieves near-baseline performance, indicating minimal impact on generation quality.

5 Related Work

Repetition in neural text generation has been widely studied. Xu et al. (2022) first demonstrated the "self-reinforcement effect," where models increasingly favor repeating prior sentences—a stark divergence from human writing patterns.

Research attributes this behavior to architectural factors. Hiraoka and Inui (2024) identified neurons that activate cyclically during repetition, akin to in-context learning circuits. Meanwhile, Hernandez et al. (2022) showed repeated training sequences impair pattern recognition mechanisms, pushing models toward memorization. These studies collec-

tively suggest repetition stems from an imbalance between memorization and generalization.

Mitigation strategies focus on two approaches: decoding constraints like nucleus sampling (Holtzman et al., 2020) or repetition penalties (Zhu et al., 2023), which suppress redundant outputs during generation, and training interventions such as Xu et al. (2022)'s DITTO framework, which reduces repetitive biases without compromising model performance.

6 Conclusion

Our work establishes induction head toxicity as a fundamental mechanism underlying the repetition curse in LLMs. Through rigorous theoretical analysis and empirical validation, we demonstrate how specialized attention heads originally designed for in-context learning can paradoxically degrade generation quality. The proposed mitigation strategies offer practical solutions while maintaining core model capabilities. Future work should investigate training-time interventions to prevent toxicity emergence while preserving induction heads' beneficial pattern recognition abilities.

Limitations

While our study offers mechanistic insights into the repetition problem through analysis of induction head toxicity, several limitations should be acknowledged. First, our empirical evaluation is limited to decoder-only transformer architectures. It remains unclear whether our findings extend to encoder-decoder or hybrid architectures, which may exhibit repetition through different internal mechanisms.Second, our method for identifying induction heads is based on synthetic copy tasks. Although effective for controlled analysis, these tasks may not fully reflect the complexity of natural repetitive text generation. This raises concerns about whether the observed toxicity patterns persist in real-world scenarios. Third, our mitigation strategy introduces a logarithmic scaling hyperparameter that must be tuned for each model architecture. Although it performed well in our experiments, applying it to other models may require additional calibration. Moreover, selectively suppressing induction heads could inadvertently weaken in-context learning capabilities, posing a potential trade-off in practical deployment. Finally, our analysis focuses on repetition at the token level. How induction head toxicity contributes to higher-level semantic repetition remains an open question for future work.

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